

TASK 3 — LOOPS FOR DATASET PROCESSING

Complete Assessment

Objective

Understand how iteration and loops can be used for basic data processing.

Tools

- Python
- Jupyter Notebook

Deliverables

- Examples using for loops and while loops
- A program that processes a list of numerical values
- A program that calculates summary statistics using loops

Implementation

1. Basic for loop

```
numbers = [10, 20, 30, 40, 50]

for number in numbers:
    print(number)
```

2. for loop with range()

```
for i in range(1, 11):
    print(i)
```

3. range() with step size

```
for i in range(0, 21, 2):
    print(i)
```

4. Process a list and calculate squares

```
numbers = [1, 2, 3, 4, 5]
squares = []

for number in numbers:
    squares.append(number ** 2)

print("Numbers:", numbers)
print("Squares:", squares)
```

5. Iterate through a dictionary

```
student = {
    "name": "Abhishek",
    "marks": 86,
    "department": "CSE"
}

for key, value in student.items():
    print(key, ":", value)
```

6. while loop

```
count = 1

while count <= 5:
    print("Count:", count)
    count += 1
```

7. while loop with accumulator

```
number = 1
total = 0

while number <= 10:
    total += number
    number += 1

print("Sum from 1 to 10:", total)
```

8. break statement

```
for number in range(1, 11):
    if number == 6:
        break
    print(number)
```

9. continue statement

```
for number in range(1, 11):
    if number % 2 == 0:
        continue
    print("Odd number:", number)
```

10. Numerical dataset processing

```
values = [12, 25, 8, 42, 17, 30, 5, 64]

total = 0
large_values = []

for value in values:
    total += value
    if value >= 30:
        large_values.append(value)

average = total / len(values)

print("Total:", total)
print("Average:", round(average, 2))
print("Values >= 30:", large_values)
```

11. Summary statistics using loops

```
values = [12, 25, 8, 42, 17, 30, 5, 64]

total = 0
minimum = values[0]
maximum = values[0]

for value in values:
    total += value
    if value < minimum:
        minimum = value
    if value > maximum:
        maximum = value

print("Count:", len(values))
print("Total:", total)
print("Average:", round(total / len(values), 2))
print("Minimum:", minimum)
print("Maximum:", maximum)
```

12. Student Marks Dataset processing

```
import pandas as pd

students = pd.DataFrame({
    "Name": ["Aman", "Priya", "Rahul", "Neha", "Ravi",
            "Anjali", "Karan", "Sneha", "Vivek", "Pooja"],
    "Marks": [86, 68, 45, 35, 72, 94, 59, 77, 39, 63]
})

marks = students["Marks"].tolist()
total = 0
passed = 0
failed = 0
```

```

for mark in marks:
    total += mark
    if mark >= 40:
        passed += 1
    else:
        failed += 1

print("Total:", total)
print("Average:", round(total / len(marks), 2))
print("Passed:", passed)
print("Failed:", failed)

```

13. Student grade calculation

```

def grade(mark):
    if mark >= 90:
        return "A+"
    elif mark >= 80:
        return "A"
    elif mark >= 70:
        return "B"
    elif mark >= 60:
        return "C"
    elif mark >= 40:
        return "D"
    return "F"

for _, row in students.iterrows():
    print(row["Name"], "->", grade(row["Marks"]))

```

14. Sales processing

```

sales = [
    {"Product": "Laptop", "Units": 12, "Price": 55000},
    {"Product": "Phone", "Units": 25, "Price": 22000},
    {"Product": "Tablet", "Units": 18, "Price": 18000},
    {"Product": "Headphones", "Units": 40, "Price": 2500},
    {"Product": "Keyboard", "Units": 30, "Price": 1500},
    {"Product": "Mouse", "Units": 45, "Price": 900}
]

total_revenue = 0

for item in sales:
    revenue = item["Units"] * item["Price"]
    total_revenue += revenue
    print(item["Product"], "Revenue:", revenue)

print("Total Revenue:", total_revenue)

```

15. Find best-selling product

```

best = sales[0]

for item in sales:
    if item["Units"] > best["Units"]:
        best = item

print("Best-selling product:", best["Product"])
print("Units sold:", best["Units"])

```

16. Nested loops

```

for i in range(1, 4):
    for j in range(1, 6):
        print(i, "x", j, "=", i * j)
    print()

```

17. Mini Project — Dataset Processing Report

```

def dataset_report(values):
    total = 0
    minimum = values[0]
    maximum = values[0]
    even_count = 0
    odd_count = 0

    for value in values:
        total += value

```

```
        if value < minimum:
            minimum = value
        if value > maximum:
            maximum = value

        if value % 2 == 0:
            even_count += 1
        else:
            odd_count += 1

    return {
        "Count": len(values),
        "Total": total,
        "Average": round(total / len(values), 2),
        "Minimum": minimum,
        "Maximum": maximum,
        "Even Count": even_count,
        "Odd Count": odd_count
    }

report = dataset_report([15, 22, 31, 44, 18, 27, 50, 63, 72, 9])
print(report)
```

Interview Questions & Answers

1. What is the difference between a for loop and a while loop?

A for loop is generally used to iterate over a sequence or a known range. A while loop repeats as long as its condition remains true.

2. What does range() do?

range() generates a sequence of integers commonly used for controlled iteration in a for loop.

3. What is an accumulator?

An accumulator is a variable that stores a running result, such as a total, count, or sum.

4. How do you iterate through a dictionary?

Use methods such as .items(), .keys(), or .values().

5. What does break do?

break immediately terminates the current loop.

6. What does continue do?

continue skips the remaining statements in the current iteration and moves to the next iteration.

7. Why are loops useful in data processing?

They automate repetitive calculations such as totals, counts, filtering, comparisons, and record processing.

8. Why are vectorized operations generally preferred over loops in data science?

Pandas and NumPy can perform many operations in optimized code, which is often faster and more concise than Python-level loops.

9. What can cause an infinite while loop?

If the loop condition never becomes false, the loop can continue indefinitely. Updating the loop-control variable helps prevent this.

10. Can loops be nested?

Yes. A loop can be placed inside another loop; this is called a nested loop.